Dear Editors,

Please consider our Article, “Input matched multisignal colocalization clarifies regulatory assignments at primary biliary cholangitis loci”, for npj Genomic Medicine.

Regulatory gene prioritization can change when colocalization inputs and signal models differ. Using primary biliary cholangitis as a fixed disease context, we assembled a traceable PBC-wide screen and evaluated a protocol-frozen high-information subset. The central contribution is an empirical reliability assessment: exact input matching separates support-set and disease-statistic changes from residual bidirectional reclassification, while signal-pair coexistence and covariate sensitivity limit gene-level interpretation.

Truth-known simulations reveal both reduced false sharing and reduced shared-signal recovery, with credible-set discovery reported separately. External disease-study and linked chromatin evidence strengthen the IL12RB2 association within explicit independence and mediation limits. Donor-level liver expression establishes detectability without corrected disease enrichment. This combination is intended to inform defensible functional prioritization; it does not propose a universally superior new algorithm, a newly discovered PBC gene or an established treatment mechanism.

The manuscript is original and is not under consideration elsewhere. The author-confirmed declarations are no specific funding, no competing interests and no additional ethics approval for this secondary analysis of public de-identified data. Generative-AI assistance is disclosed in Methods. Code, aggregate evidence and source provenance are available in the public repository; third-party individual-level inputs are not redistributed.

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Sincerely,
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